Project: Worked examples
Title: Enclosure heat dissipation and fan sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the heat load of a control enclosure, its natural temperature rise and the airflow for a target rise. It also gives the fan duty point and the cooling-unit capacity needed when the limit cannot be met.

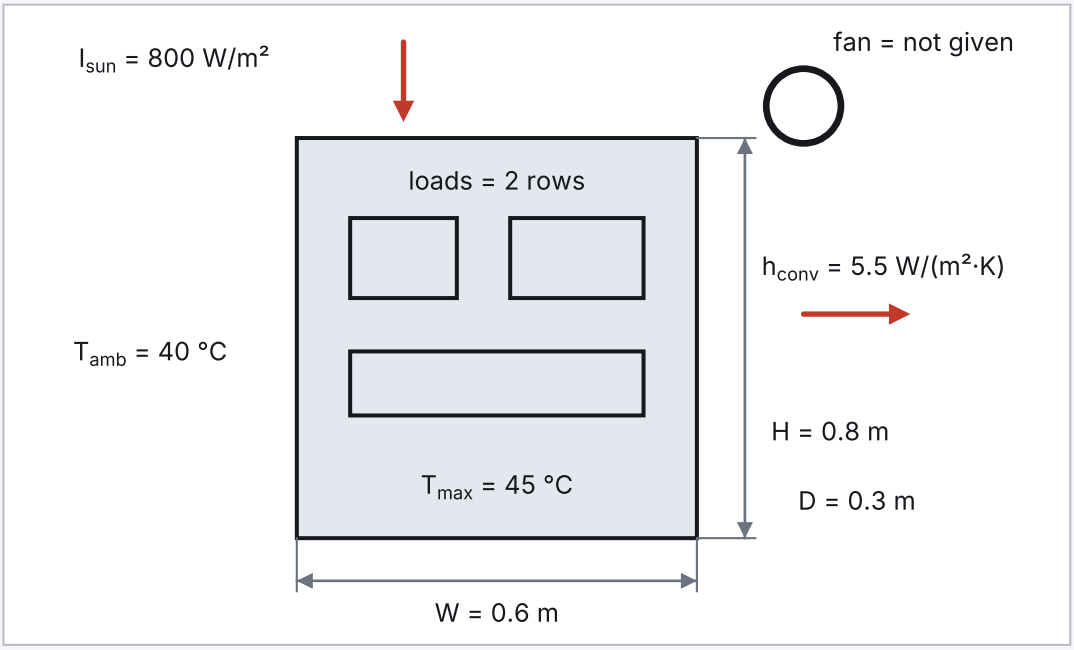


Figure 1 — An enclosure W by H by D holding the heat loads as listed, losing heat through its exposed faces to the ambient, with a fan and the sun on it. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Heat dissipated inside the enclosure, plus any solar gain	$P = \Sigma P_i n_i d_i (+ \alpha I A_{sun})$
Effective cooling surface, each face weighted by how freely it sheds heat	$A_{eff} = \Sigma f_{face} \times A_{face}$
Temperature rise with natural cooling, one coefficient over the exposed faces (the IEC TR 60890 tables are not used)	$\Delta T = P/(h_{conv} \cdot A_{eff})$
Air flow a fan must move to hold the target rise	$V = P/(\rho\, c_p\, \Delta T_{target})$, with $\rho = p(z)/(287.055\, T)$ and c_p = 1006 J/(kg·K)
Fan duty point, where the fan curve meets the system resistance (found by bisection)	$p_{fan}(V) = p_{sys}\,(V/V_{ref})^2$
Cooling unit capacity, when the surfaces alone cannot reject the heat	$P_{cool} = P - h_{conv} \cdot A_{eff} \cdot (T_{max} - T_{amb})$; with T_{max} at or below T_{amb} the term is a gain, $P_{cool} = P + h_{conv} \cdot A_{eff} \cdot (T_{amb} - T_{max})$, and no ventilation airflow or fan duty is given

WHAT IT CALCULATES	RELATION
Sealed enclosure (sealing temperature T_{seal} entered, no breather)	$\Delta p = p_{site} \cdot \lvert T_{int}/T_{seal} - 1 \rvert$, the pressure difference across the walls when the air warms or cools from the sealing temperature at constant volume, in absolute temperatures (a rise when $T_{int} > T_{seal}$, a suction when below); a breather or relief element limits it, $dp_{seal} = \min(\Delta p, dp_{relief})$; the force on a cover of area A_{cover} is $F_{seal} = dp_{seal} \cdot A_{cover}$

Basis: A method of temperature-rise verification of low-voltage switchgear and controlgear assemblies by calculation[1]; Verification of temperature rise (edition to be checked)[2](cl. 10.10).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Heat loads	2 row(s)		spec: Panel schedule PNL-09	any
W	Enclosure width	0.6	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
H	Enclosure height	0.8	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
D	Enclosure depth	0.3	m	spec: Enclosure datasheet AE-1380	0.1 … 5 m
f_{top}	Top exposure (1 free, 0 covered)	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{front}	Front exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{rear}	Rear exposure (0 = wall-mounted)	1		assumption: Free-standing.	0 … 1
f_{left}	Left side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{right}	Right side exposure	1		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
f_{bottom}	Bottom exposure (0 = on the floor)	0		assumption: Free-standing kiosk, bottom on a plinth.	0 … 1
A_{sun}	Sunlit area	0.48	m²	assumption: One 600 × 800 face in the sun.	0 … 100 m²
α_{sun}	Surface absorptivity	0.5		assumption: Light-grey paint.	0.1 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
h_{conv}	Overall heat-transfer coefficient of the enclosure surface (editable reference value)	5.5	W/(m²·K)	reference: Reference record: painted steel enclosure, still air, 5.5 W/m²K	3 … 8 W/(m²·K)
P_{add}	Additional heat load (linked from the load-budget dissipated heat or another calculation)	0	kW	assumption: No further heat load linked into this kiosk.	0 … 100 kW
I_{sun}	Solar irradiance on the enclosure (0 = indoors)	800	W/m²	spec: Site conditions SC-5: 800 W/m² on the sunlit side	0 … 1200 W/m²
T_{amb}	Ambient temperature	40	°C	spec: Site conditions SC-5	-20 … 55 °C
altitude	Site altitude (air density)	0	m	assumption: Free-standing kiosk, bottom on a plinth.	0 … 4000 m
T_{max}	Maximum internal temperature	45	°C	spec: Site conditions SC-5	20 … 80 °C
p_{sys}	System resistance (filters, grilles) at the reference airflow	20	Pa	assumption: No filter fan on this kiosk (it is cooled by a cooling unit); template default kept.	0 … 500 Pa
V_{ref}	Reference airflow for the system resistance	100	m³/h	assumption: No filter fan on this kiosk (it is cooled by a cooling unit); template default kept.	1 … 100000 m³/h

TABLE ROW	VALUES
loads[1]	name = PLC and I/O; P = 80 W; qty = 1; duty = 1
loads[2]	name = drives; P = 55 W; qty = 4; duty = 1

3. Assumptions

A1. Heat leaves the enclosure through its exposed faces at one overall coefficient h_{conv}. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A2. The coefficient covers natural convection and radiation to surroundings at the ambient temperature. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A3. Covered faces dissipate nothing. (Plain exposed-area method of manufacturer practice, not an IEC TR 60890 verification; h_{conv} is a reference value the user may edit.)
A4. Forced ventilation removes the full heat load with air entering at ambient and leaving at the target rise. (Manufacturer fan-sizing practice.)
A5. Natural dissipation is neglected for the airflow, which is conservative. (Manufacturer fan-sizing practice.)

A6. A sealed enclosure (T_{seal} entered) is a closed, rigid, leak-free volume of dry air at the site pressure when closed. (Ideal gas at constant volume (derived).)
A7. The pressure difference follows the ideal-gas relation at constant volume, from the sealing temperature to T_{int}. (Ideal gas at constant volume (derived).)
A8. T_{int} is the internal temperature with natural dissipation, without a cooling unit. (Ideal gas at constant volume (derived).)
A9. A breather, a seal that leaks or a flexing wall reduces the difference, so the result is an upper bound. (Ideal gas at constant volume (derived).)
A10. A pressure-compensation element is modelled only as a cap at its cracking pressure, with no flow resistance. (Ideal gas at constant volume (derived).)
A11. The site pressure is the standard atmosphere at the altitude. (Ideal gas at constant volume (derived).)
A12. The fan curve is linear between the entered points. (Duty-point idealisation.)
A13. The system resistance scales with airflow squared. (Duty-point idealisation.)
A14. f_{top} (Top exposure (1 free, 0 covered)) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A15. f_{front} (Front exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A16. f_{rear} (Rear exposure (0 = wall-mounted)) is taken as 1. (Free-standing.)
A17. f_{left} (Left side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A18. f_{right} (Right side exposure) is taken as 1. (Free-standing kiosk, bottom on a plinth.)
A19. f_{bottom} (Bottom exposure (0 = on the floor)) is taken as 0. (Free-standing kiosk, bottom on a plinth.)
A20. A_{sun} (Sunlit area) is taken as 0.48 m². (One 600 × 800 face in the sun.)
A21. α_{sun} (Surface absorptivity) is taken as 0.5. (Light-grey paint.)
A22. P_{add} (Additional heat load (linked from the load-budget dissipated heat or another calculation)) is taken as 0 kW. (No further heat load linked into this kiosk.)
A23. altitude (Site altitude (air density)) is taken as 0 m. (Free-standing kiosk, bottom on a plinth.)
A24. p_{sys} (System resistance (filters, grilles) at the reference airflow) is taken as 20 Pa. (No filter fan on this kiosk (it is cooled by a cooling unit); template default kept.)
A25. V_{ref} (Reference airflow for the system resistance) is taken as 100 m³/h. (No filter fan on this kiosk (it is cooled by a cooling unit); template default kept.)
A26. The effective surface is the plain exposed-area method of manufacturer practice: each face area weighted by how freely it sheds heat, one overall coefficient h. This is not a verification of temperature rise to IEC TR 60890 or BS EN 61439-1 cl. 10.10 (neither the surface factors nor the standard's temperature-rise correlation is used). (Stated on every run so the result is not read as an assembly verification.)

4. Working

[18] $P_{loads} = \Sigma\ P_i \times n_i \times d_i$
$= 2\ \text{item(s)}$
$= 0.3\ \text{kW}$
Internal heat load

[19] $P_{solar} = \alpha_{sun} * I_{sun} * A_{sun}$
$= 0.5 \times 800\ \text{W/m}^2 \times 0.48\ \text{m}^2$
$= 0.192\ \text{kW}$
Solar gain

[20] $P_{total} = P_{loads} + P_{add} + P_{solar}$
$= 0.3\ \text{kW} + 0\ \text{kW} + 0.192\ \text{kW}$
$= 0.492\ \text{kW}$
Total heat to remove

[21] $A_{eff} = f_{top} * W * D + f_{bottom} * W * D + f_{front} * W * H + f_{rear} * W * H + f_{left} * H * D + f_{right} * H * D$ [1](verification)
$= 1 \times 0.6\ \text{m} \times 0.3\ \text{m} + 0 \times 0.6\ \text{m} \times 0.3\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.6\ \text{m} \times 0.8\ \text{m} + 1 \times 0.8\ \text{m} \times 0.3\ \text{m} + 1 \times 0.8\ \text{m} \times 0.3\ \text{m}$
$= 1.62\ \text{m}^2$
Effective dissipating surface

[22] $dT_{nat} = P_{total} / (h_{conv} * A_{eff})$
$= 0.492\ kW / (5.5\ W/(m^2 \cdot K) \times 1.62\ m^2)$
$= 55.219\ K$
Temperature rise with natural dissipation
[23] $T_{int} = T_{amb} + dT_{nat}$
$= 40\ degC + 55.219\ K$
$= 95.219\ degC$
Internal temperature with natural dissipation
[24] $dT_{allow} = T_{max} - T_{amb}$
$= 45\ degC - 40\ degC$
$= 5\ K$
Allowed rise
[25] $u_{nat} = dT_{nat} / dT_{allow}$
$= 55.219\ K / 5\ K$
$= 11.044$
Natural dissipation utilisation
[26] $P_{nat,max} = h_{conv} * A_{eff} * dT_{allow}$
$= 5.5\ W/(m^2 \cdot K) \times 1.62\ m^2 \times 5\ K$
$= 0.04455\ kW$
Heat the enclosure dissipates naturally at the limit
[27] $P_{cool} = max(0\ W, P_{total} - P_{nat,max})$
$= max(0\ W, 0.492\ kW - 0.04455\ kW)$
$= 0.44745\ kW$
Cooling-unit capacity needed if ventilation is not used
[28] $p_{site} = 101325\ (1 - 2.25577e\text{-}5\ z)^{5.2559}$
$= z = 0\ m$
$= 101.32\ kPa$
Site pressure
[29] $\rho_{air} = p_{site} / (287.055\ J/(kg*K) * T_{amb})$
$= 101.32\ kPa / (287.055\ J/(kg*K) \times 40\ degC)$
$= 1.1272\ kg/m^3$
Air density at ambient and altitude
[30] $V_{req} = P_{total} / (\rho_{air} * 1006\ J/(kg*K) * dT_{allow})$
$= 0.492\ kW / (1.1272\ kg/m^3 \times 1006\ J/(kg*K) \times 5\ K)$
$= 312.39\ m^3/h$
Airflow for forced ventilation at the allowed rise (natural dissipation neglected)
[31] $V_{req,net} = max(0\ m^3/h, (P_{total} - P_{nat,max}) / (\rho_{air} * 1006\ J/(kg*K) * dT_{allow}))$
$= max(0\ m^3/h, (0.492\ kW - 0.04455\ kW) / (1.1272\ kg/m^3 \times 1006\ J/(kg*K) \times 5\ K))$
$= 284.11\ m^3/h$
Airflow crediting natural dissipation

5. Results

SYMBOL	VALUE	DESCRIPTION
P_{total}	0.4920 kW	Total heat
P_{cool}	0.4474 kW	Cooling unit needed
$P_{nat,max}$	0.04455 kW	Natural dissipation at the limit
T_{max}	45 °C	Internal temperature limit
dT_{nat}	55.22 K	Natural temperature rise (governing)
T_{int}	95.22 °C	Internal temperature (natural)
V_{req}	312.4 m³/h	Required airflow

SYMBOL	VALUE	DESCRIPTION
P_{loads}	0.3000 kW	Internal heat load
P_{solar}	0.1920 kW	Solar gain
A_{eff}	1.620 m²	Effective surface
ρ_{air}	1.127 kg/m³	Air density
$V_{req,net}$	284.1 m³/h	Required airflow (net)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Natural temperature rise against the allowed rise	55.22 K	5 K	11.05	

7. Applicability

WARNING T_{max}: natural dissipation holds 44.55 W of 492 W. Ventilate, or fit a 447.4 W cooling unit. [cooling-unit]

8. Verification

Template elec.enclosure-heat@1.7.1 Content hash c419a03dc9212448 Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. International Electrotechnical Commission. A Method of Temperature-Rise Verification of Low-Voltage Switchgear and Controlgear Assemblies by Calculation. IEC TR 60890. IEC.
2. British Standards Institution. Low-Voltage Switchgear and Controlgear Assemblies—Part 1: General Rules. BS EN 61439-1. BSI.